

Case Study: Effectiveness and Scale-Consistency of Qwen3-VL on Identifying Low-Level Perceptual Features

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Abstract

Multimodal large language models (MLLMs) have shown strong performance on a wide range of vision-language tasks, but their ability to capture low-level perceptual features remains under-explored. In this work, we present a case study evaluating the effectiveness and scale-consistency of Qwen3-VL on identifying primitive perceptual signals. We design three procedurally generated tasks—color recognition, location recognition, and shape recognition—that isolate low-level visual features and provide precise ground truth. We further evaluate the impact of different output scales, including Likert scales, symmetric ranges, and continuous intervals, to examine whether MLLMs produce consistent outputs across scales. Our results show that performance improves with model size and stabilizes beyond the 8B parameters. However, smaller models exhibit strong scale sensitivity, and no single output scale performs best on all tasks. Additionally, we observe that MLLMs underutilize continuous output spaces, frequently collapsing to a small set of discrete values. These findings highlight fundamental limitations in both perceptual grounding and numerical reasoning in MLLMs.

1 Introduction

Multimodal large language models (MLLMs) have recently achieved impressive results in a wide range of tasks that combine vision and language. Despite this progress, there remain open questions about their ability to reason over low-level perceptual features, such as color, spatial location, and geometric shape. Most existing benchmarks focus on high-level semantic understanding and rely heavily on images collected from the internet, making it difficult to isolate the model’s true perceptual capabilities.

In this work, our objective is to evaluate MLLMs from a different perspective by focusing entirely on primitive perceptual signals. Instead of relying on natural images, we procedurally generate task instances with precise control over the underlying ground truth. This allows us to construct evaluation tasks that are independent of semantic biases and di-

rectly measure how well models can capture fine-grained visual properties.

Another important but often overlooked factor in evaluating language models is the choice of output scale. Previous work has shown that large language models can be sensitive to the format and range of output, especially when used as judges or human proxies. However, this issue has been studied primarily in language-only settings. It remains unclear whether similar scale inconsistencies exist in multimodal settings, particularly when models are required to map perceptual inputs to numerical outputs.

To address these gaps, we conducted a systematic study on Qwen3-VL, evaluating both its perceptual accuracy and its consistency on different output scales. We design three tasks—color recognition, location recognition, and shape recognition—and test the model on multiple scales, including Likert scales, symmetric ranges, and continuous intervals. Our experiments aim to answer two key questions: (1) how well can MLLMs identify low-level perceptual features, and (2) how consistent are their outputs across different numerical scales?

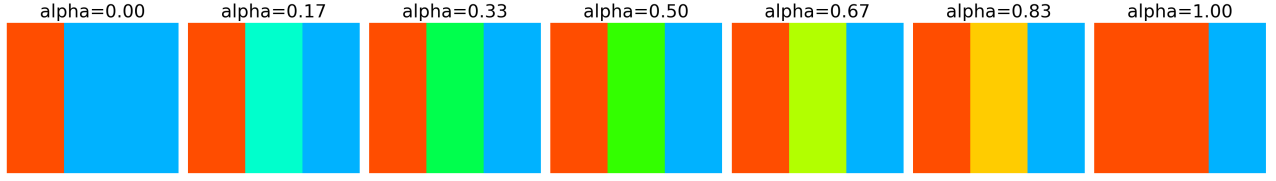
Our results show that, while performance improves with model size, smaller models are highly sensitive to the choice of output scale. Furthermore, even large models exhibit limitations in utilizing continuous output spaces, suggesting that their numerical reasoning remains fundamentally discrete. These findings provide new insights into the perceptual and numerical limitations of MLLMs and highlight the importance of carefully designing evaluation protocols.

2 Related Work

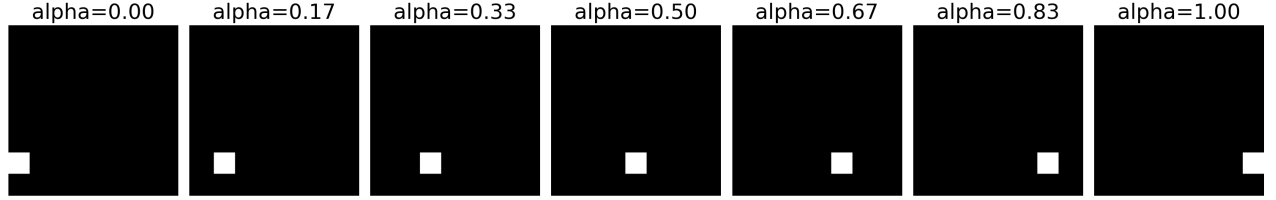
2.1 Perceptual Limitations in MLLMs

Many existing benchmarks have exposed the perceptual limitations of MLLMs. Specifically, MLLMs are criticized for being poor in physical world grounding [Chow *et al.*, 2025], motion estimation [Hong *et al.*, 2025], visual search in high-resolution images [Wu and Xie, 2023], counting [Guo *et al.*, 2025], and have strong language bias [Liu *et al.*, 2025].

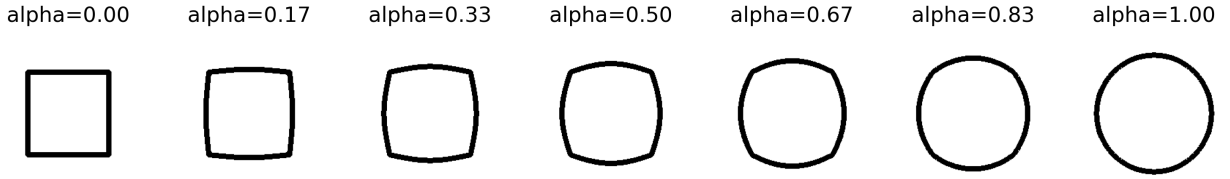
Unlike previous work that collects images from the internet and has tasks that are heavily dependent on the semantic information, in this work we procedurally generate the evaluation tasks and focus entirely on very low-level visual features. Additionally, we also consider the effect of different



(a) Color recognition task instances. Images reshaped for better visualization. Best viewed in color.



(b) Location recognition task instances



(c) Shape recognition task instances

Figure 1: Low-level perceptual tasks used.

scale to make sure the output format is not the bottleneck of MLLMs.

2.2 Scale Consistency in LLMs

Using LLMs to give responses as human proxies [Seshadri *et al.*, 2026] and as judges [Gu *et al.*, 2025], many works have investigated whether LLMs can provide subjective output reliably. Benchmarks [Bai *et al.*, 2024; Lambert *et al.*, 2024] are created to evaluate how well LLMs are at assigning process based rewards. Many works have shown that the output scale [Lee *et al.*, 2025; Li *et al.*, 2026] has a noticeable influence on the accuracy of the LLMs and that the output is not always consistent. This is potentially due to the tokenizers not carrying the semantics of numbers [Singh and Strouse, 2024]. Other work has looked at how to query the output distribution for a more LLM-specific, fine-grained output scale [Wang *et al.*, 2025].

In this work, we extend the evaluation from language only to multi-modal inputs. We also test on multiple scales to evaluate inter-scale consistency.

3 Experiment Setup

3.1 Tasks

In this work, we designed three low-level perception tasks to evaluate the MLLM’s capability of handling very primitive

perceptual information. All task instances can be procedurally generated, such that we can trivially scale up the experiment and have an accurate comparison between the model’s output and the ground truth. For all task instances, the answer is a number between 0 and 1 (inclusive).

Color Recognition

In this task, a pair of complementary colors are randomly generated as anchor colors. That is, in the HSV color space,

$$\text{color}_1 = (x, 1, 1) \quad \text{color}_2 = (y, 1, 1) \quad |x - y| = 180^\circ$$

Then, we randomly generated the ground truth $\alpha \in [0, 1]$ and constructed the third blended color by

$$\text{color}_{\text{blended}} = (\alpha x + (1 - \alpha)y, 1, 1)$$

Note that we only alter the hue of the colors while keeping the saturation and intensity at the maximum. This has been shown to be closest to human perception of color [Vadivel *et al.*, 2005].

The MLLM is prompted with the two anchor colors and the blended color (each in the form of an 256×256 image), and the task is to identify how close the blended color is to the anchor colors. Examples of the color recognition task are shown in Figure 1a.

Location Recognition

In this task, we randomly generate a black or white image with a small square on it (Figure 1b). The location of the

small square is generated randomly, and its horizontal location is used as the ground truth. The MLLM is asked to estimate whether the square is to the left or right of the image.

Shape Recognition

In this task, we first sample a pair of distinct shapes from a set of pre-defined shapes: circle, equilateral triangle, square, and pentagon. Then, we randomly generated the ground truth $\alpha \in [0, 1]$ and constructed the morphed shape using radial interpolation.

Specifically, we can express all the shapes in polar coordinates, where a circle is defined as

$$r_{\text{circle}}(\theta) = 1$$

And a regular n-gon is

$$r_n(\theta) = \frac{1}{\cos\left(\theta \bmod \frac{2\pi}{n} - \frac{\pi}{n}\right)}$$

Then given a pair of shapes r^1 and r^2 , together with the ground truth α , we can create a morphed shape by

$$r_{\text{morphed}}(\theta) = \alpha r^1(\theta) + (1 - \alpha)r^2(\theta)$$

The MLLM is prompted with the morphed image and the task is to identify how close the morphed image is to the shapes. Figure 1c shows an example of morphing between a square and a circle.

3.2 Scales

We include multiple scale as the output format for the MLLM to test for scale-consistency and to minimize the impact of using a particular scale. In particular, we include the following:

- Likert scale that is commonly used in human behavior studies [Likert, 1932]: 5 point, 7 point, and 10 point Likert scales
- Survey style scale: 0 to 100
- Symmetric scales: -3 to 3 , -5 to 5 , and -100 to 100
- Continuous output commonly used for procedure reward models [Zheng *et al.*, 2025]: 0 to 1 or -1 to 1 .

For clarity, we use $[x..y]$ to denote all integers in the range of x to y inclusive, and use (x, y) to denote all floating point numbers in the range of x and y inclusive.

3.3 Evaluation Process

We use Qwen3-VL-Instruct [Bai *et al.*, 2025] as the MLLM for this experiment. It is chosen because it is open-source and has multiple sizes for us to evaluate how the size of the VLLM affects the results. Specifically, we use all four dense models: 2B, 4B, 8B, 32B. We keep the sampling temperature to the recommended 0.7 to avoid loop outputs.

For each task, we generate 200 instances. For each instance-scale pair, we query the MLLM for a number as the output (sample query provided in listing 1). The output is normalized w.r.t. the provided scale and will always be between 0 and 1. This results in $3 \times 200 \times 9 \times 4 = 21600$ tests.

Once we have all the results, we evaluate the correlation between outputs in each scale and the ground truth answer using the Pearson correlation coefficient [Pearson, 1909]. Additionally, we also evaluate the correlation between different output scales.

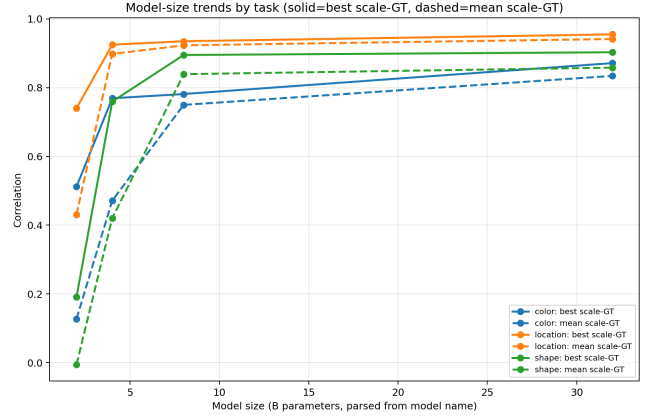


Figure 2: Output-GT correlation for each task at different model sizes

Listing 1: Sample prompting for the shape task with the 0 - 1 scale

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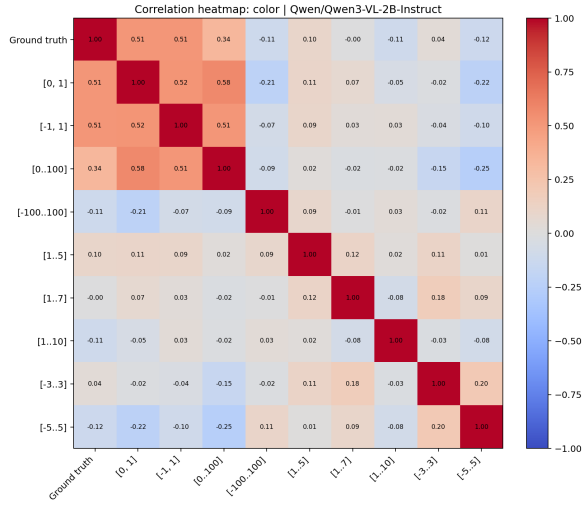
1 <system>
2 You are a precise visual reasoning
  assistant. Follow the formatting
  instructions exactly. Return only one
  XML answer block in the form <answer>
  NUMBER</answer>.
3 </system>
4 <user>
5 <image> Question: Is the morphed shape
  closer to equilateral triangle or
  circle?\nRespond with exactly a
  floating-point number in [0.0, 1.0],
  wrapped as <answer>...</answer>.\n
  Interpretation: 1 means definitely
  closer to equilateral triangle; 0
  means definitely closer to circle; 0.5
  means equally close.
6 </user>
```

4 Results

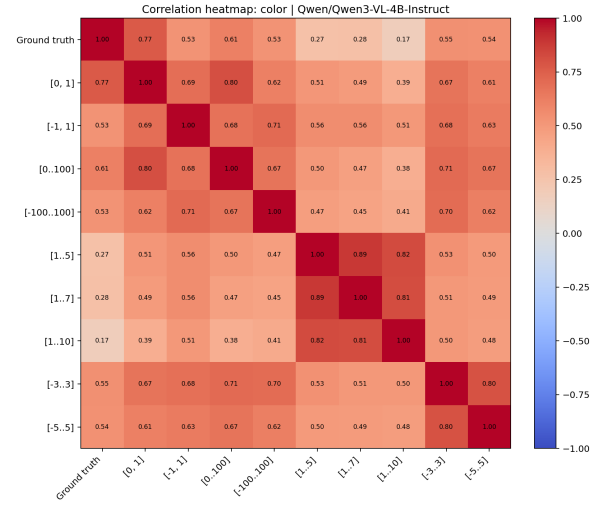
4.1 Performance Improves with Model Size

Unsurprisingly, the performance of the MLLM, both in terms of output-GT correlations and inter-scale correlations, increases as the model size increases (Figure 2). Generally, location recognition performs the best, mostly attribute to the multimodal position embedding in the Qwen3-VL family. The correlations also plateau beyond the 8B models, indicating that it is sufficient for low-level visual perception.

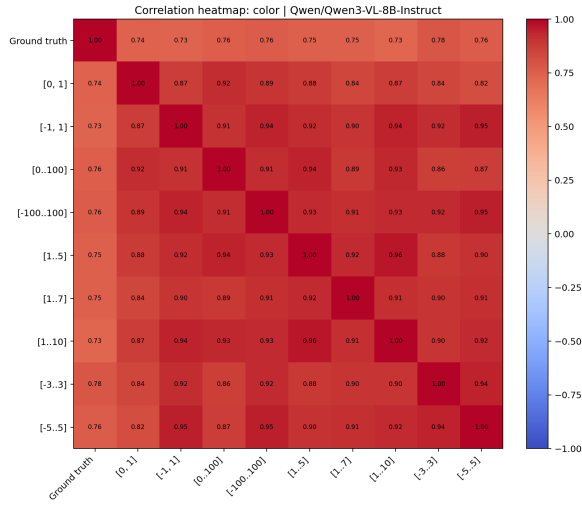
Figure 3 shows the correlation heatmap of the four models on the color recognition task. The 2B model has very poor performance, indicating that fine-grained color perception is lost during the distillation process, as well as its capability to convert between different scales. This shows that instead of forming a general impression of the answer before fitting it into the output scale, the model directly predicts the output of the input task. At the 8B level, the model is able to perform consistently across different output scales.



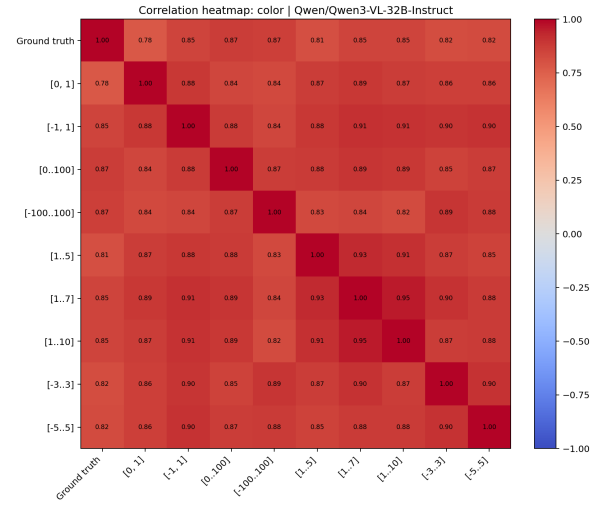
(a) Qwen3-VL-2B on the color recognition task



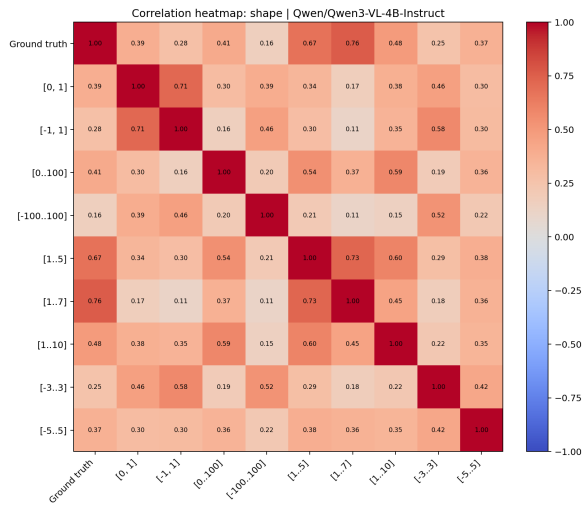
(b) Qwen3-VL-4B on the color recognition task



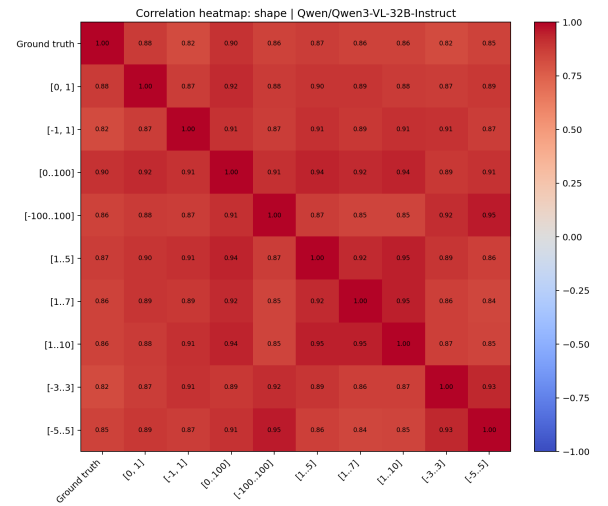
(c) Qwen3-VL-8B on the color recognition task



(d) Qwen3-VL-32B on the color recognition task



(e) Qwen3-VL-4B on the shape recognition task



(f) Qwen3-VL-32B on the shape recognition task

Figure 3: Heatmaps across different scales for the color recognition and shape recognition tasks. Note that the first column and the first row of each figure represent the ground truth. Best viewed in color.

4.2 Different Tasks Have Different Optimal Scale

In the color recognition task, the scale $[0, 1]$ achieves the best correlation between ground truth and the 4B model (Figure 3b). However, it performs poorly for the shape recognition task for the same model, and the $[1..7]$ scale is the best (Figure 3e). This shows that for small models, there may not be a universal scale that works best for all tasks.

However, on a large model, the difference is much less noticeable. This is shown between Figure 3d and Figure 3f. This indicates that larger models had more language reasoning capability and could easily convert from one scale to another.

4.3 Inter-Scale Consistency Depends on Scale Types

Clearly shown in Figures 3a, 3b, and 3e, the consistency between scales of the small models depends on the type of evaluation scales. Generally, each scale category has higher intra-category consistency than inter-category consistency. That is, 5-, 7-, 10-point Likert scales are more consistent with each other compared to w.r.t. floating point scales. Interesting, the symmetric scales (e.g., $[-3..3]$) have good consistency between themselves but not with the Likert scales. This likely suggests that MLLMs do not perceive numbers as shift invariant due to the tokenization of the extra negative sign.

4.4 Floating Point Scales are Under Utilized

One of the advantages of having a continuous output scale (e.g. $[0, 1]$) as opposed to a fixed-point Likert scale is that it is capable of generating more nuances. However, it turns out that MLLMs cannot leverage this flexibility and large output of the same numbers even when arbitrary floating point number is allowed. This behavior also does not improve as the model is larger (Figure 4). This shows that the MLLMs are fundamentally reasoning in the discrete space and their capability of generating decimal output is unreliable.

However, as the model size grows to 8B, the MLLM achieves good performance for the tasks across all scales, thus minimizing the inter-scale discrepancies.

4.5 Dense MLLM Outperforms MOE MLLM

We also found that MLLMs with the Mixture-of-Expert (MOE) architecture significantly underperform compared to their dense counterparts at the same size (32B dense vs 30B MOE). Empirically, we found that the dense models always output the answer directly, while the MOE model gives a length reasoning despite receiving the same prompt. Sometimes, the reasoning is very confined to the discrete space (e.g., first convert the color image into the name of the color then estimate how close it is to the anchor colors), limiting the performance.

5 Limitations and Future Work

In this work, we only evaluated one family of MLLM (Qwen3-VL). Given that different MLLMs have different modality fusion mechanisms [Wei *et al.*, 2025] and different variants of the attention mechanism [Team *et al.*, 2026], the conclusion may not be directly translated into other MLLMs. Additionally, due to compute constraints, we evaluated only

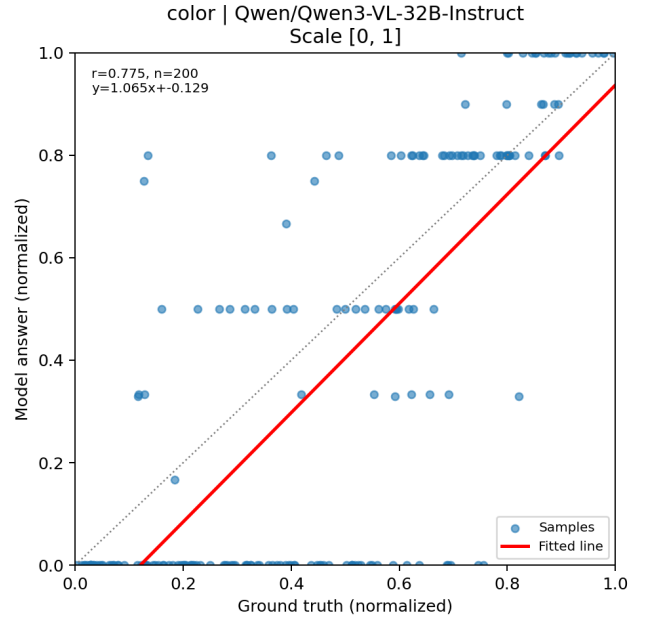


Figure 4: Qwen3-VL-32B $[0, 1]$ scale to GT on the color recognition task

up to 32B. Future work should test other MLLMs in different families and at larger scales.

Additionally, the tasks defined in this work are very primitive and do not have good coverage. Future work should investigate other factors, for example the shape of the image input, to see whether they influence the MLLMs’ performance.

6 Conclusion

In this work, we presented a case study on the effectiveness and scale-consistency of Qwen3-VL in identifying low-level perceptual features. By designing procedurally generated tasks that isolate primitive visual signals, we were able to directly evaluate the model’s perceptual capabilities without confounding semantic information.

Our experiments show that performance improves with model size and becomes relatively stable beyond 8B parameters. However, smaller models exhibit strong sensitivity to the choice of output scale, and no single scale consistently performs best across all tasks. Even for larger models, we find that continuous output spaces are not fully utilized, with predictions often collapsing to a small set of discrete values.

These findings suggest that current MLLMs still face fundamental challenges in both perceptual grounding and numerical reasoning. Improving scale-consistency and enabling better use of continuous outputs may be important directions for future research.

Acknowledgment

Coding LLM (specifically `codex-5.3` hosted on Copilot) is used to generate the code for the experiments. Gemini 3 Pro helped with some literature review and the initial draft of the introduction and conclusion sections.

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